



e-TRIEVER® Terminal Tractor



Models 2021 to OCT 2024

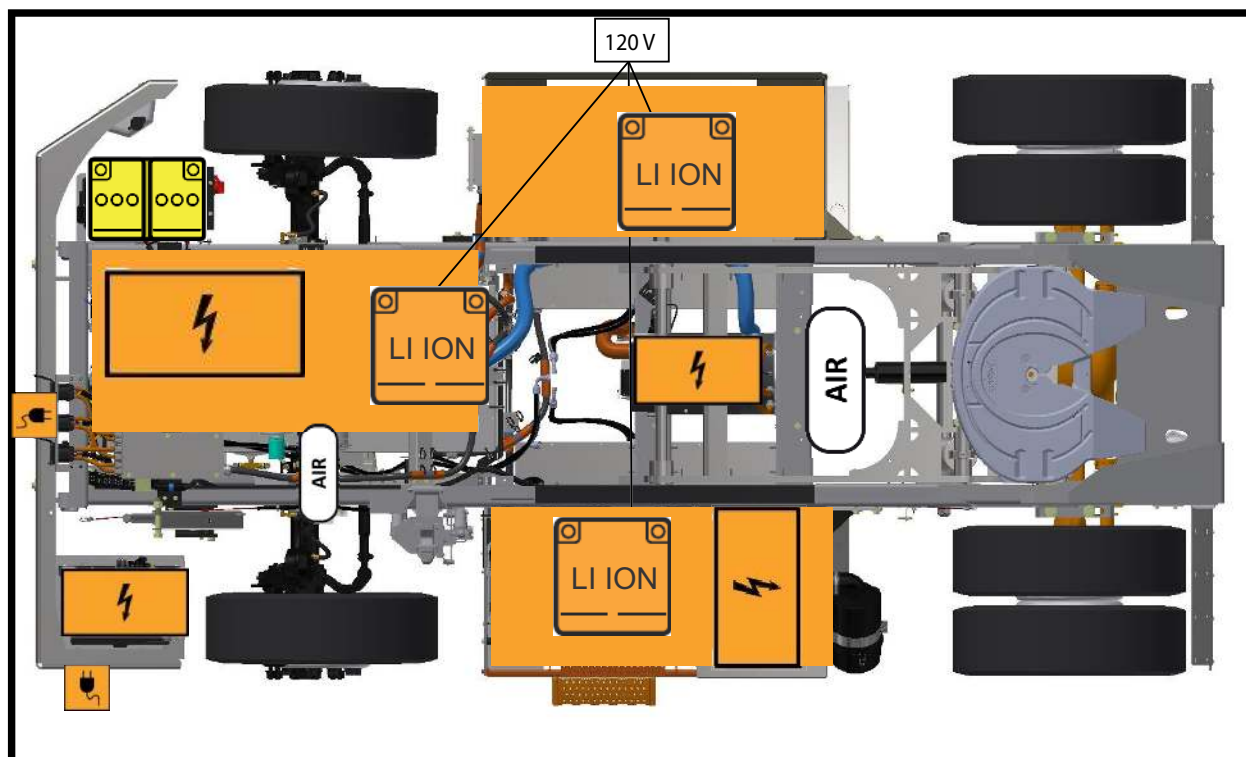
Emergency Response Guide

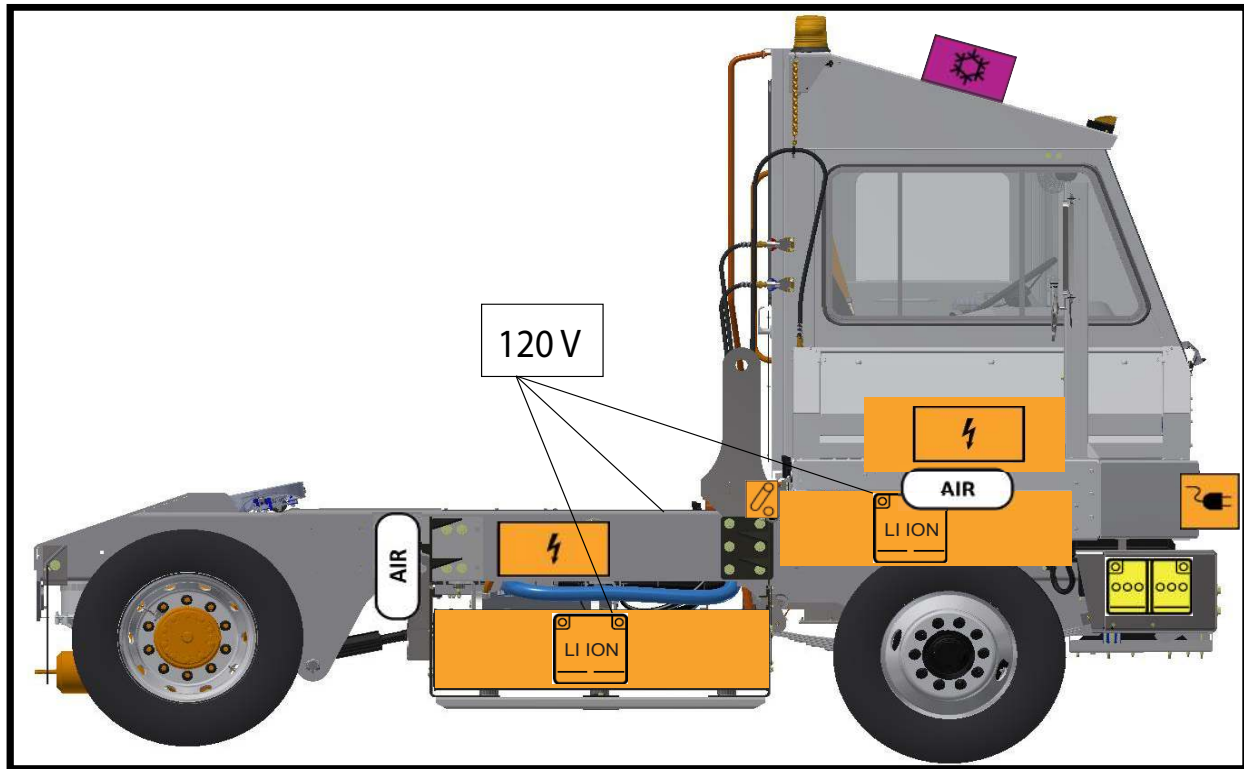


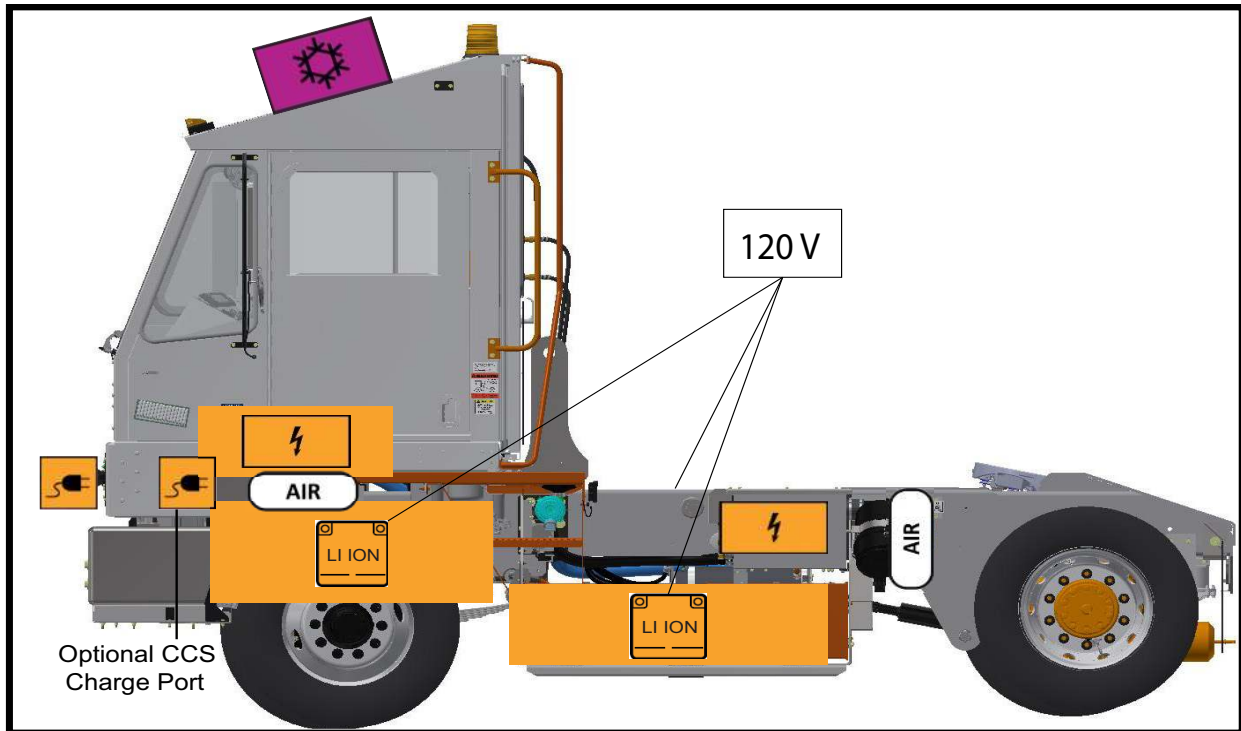


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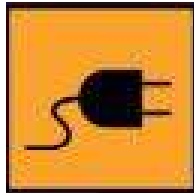
0. Emergency Response Sheet







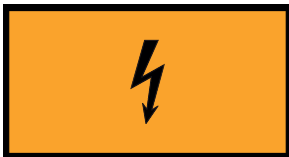
Air Conditioning Component



Charger Port



Emergency Exit Left Hand Side



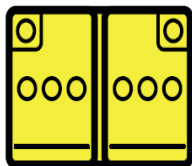
High Voltage Component



Air Tank



High Voltage Cable



Battery, Low Voltage

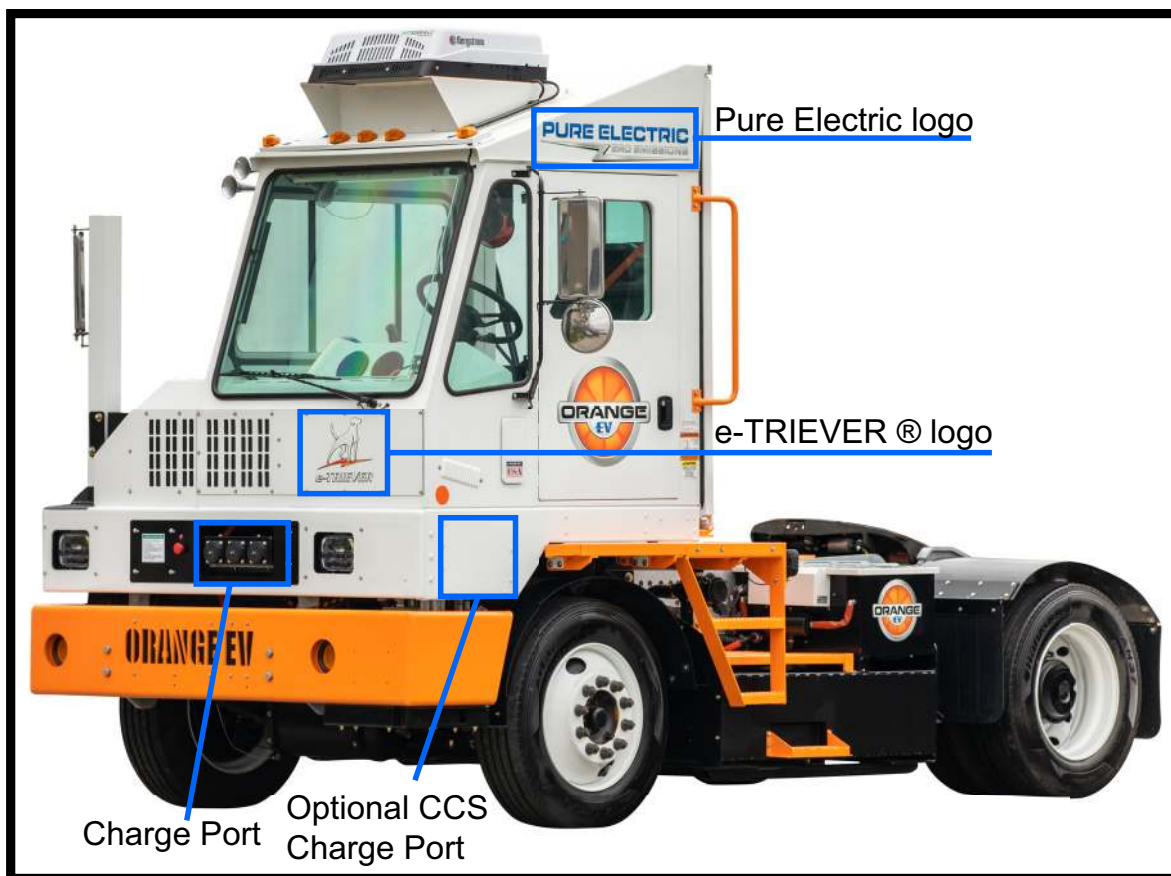


Lithium Ion Battery



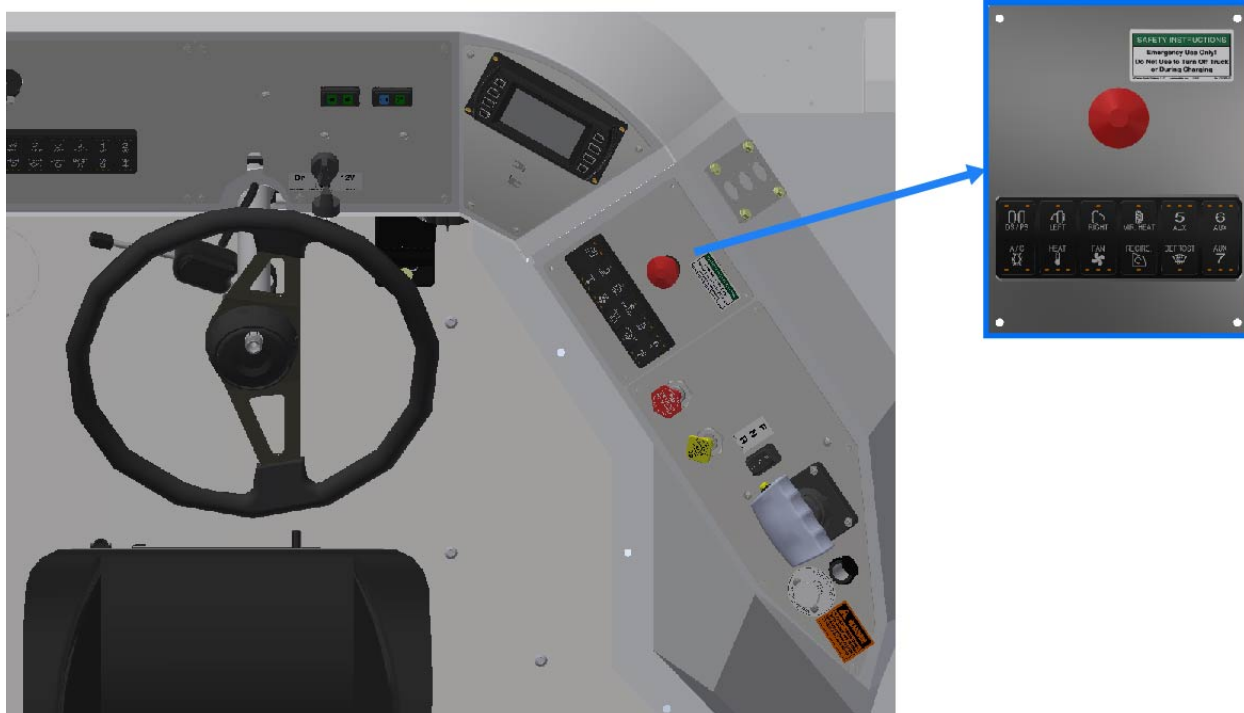
High Voltage Service Disconnect

1. Identification / Recognition



Scope

This manual covers the Orange EV e-TRIEVER® equipped with an SDHI (Safety Device Hardware Interlock). If you do not see an SDHI on the dash panel, please refer to the Orange EV e-TRIEVER Terminal Tractor 2024+.



2. Immobilization / Stabilization / Lifting



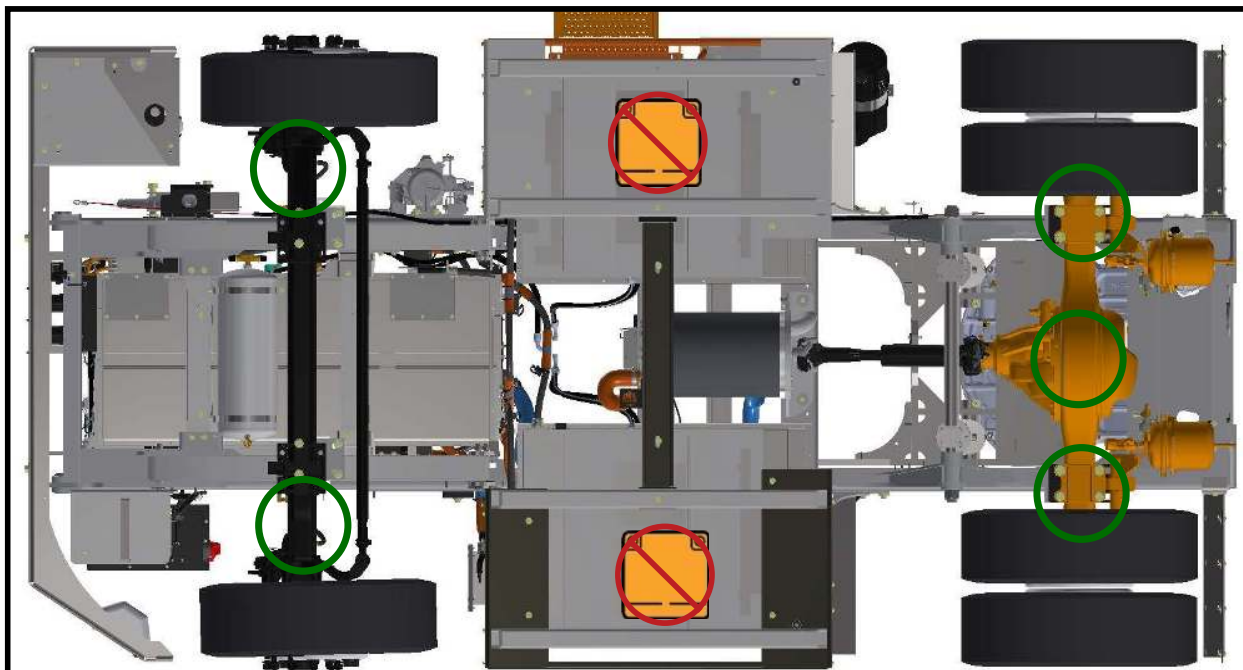
WARNING

A lack of noise from electric trucks does not indicate that movement is not possible. Silent movement remains possible until all immobilization steps are completed.

Immobilization:

1. When approaching the truck from the outside, chock front and rear wheels.
2. Set parking brake (2).
3. Set trailer brakes (1).
4. Shift truck into *N* (3).





The above image is a bottom view of the truck.

Lift truck at indicated green points on the axles.

Do not use battery boxes to lift the truck.

Do not attach lift device to the brake chamber.

3. Disable Direct Hazards / Safety Regulations

WARNING

The High Voltage System on this truck is **NOT** capable of being **LOCKED OUT**.

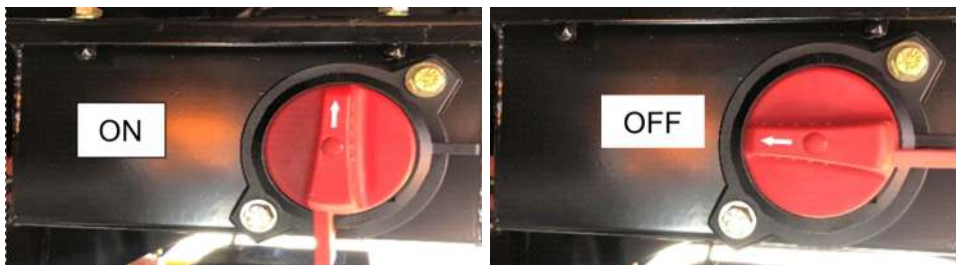
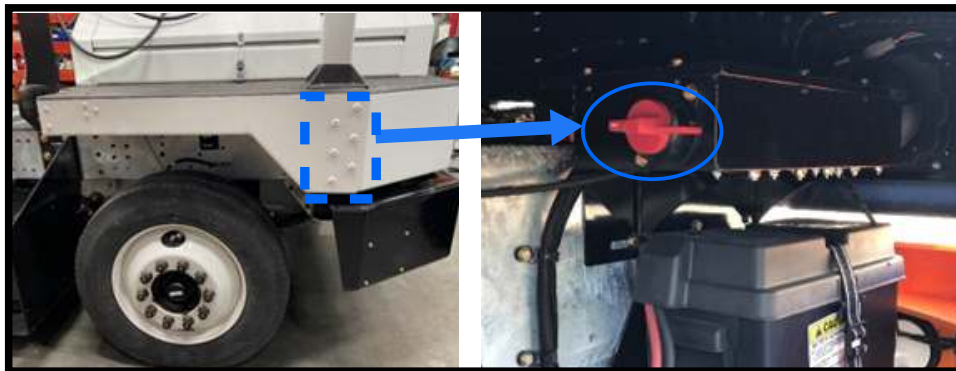
Assume the high voltage system is energized. Do not touch any high voltage cables or components.

In the event of fire, submersion, accident, or other emergency, shut down the high-voltage system by following the procedure below if it is safe to do so.

High Voltage Safety Device Hardware Interlock (SDHI) Activation

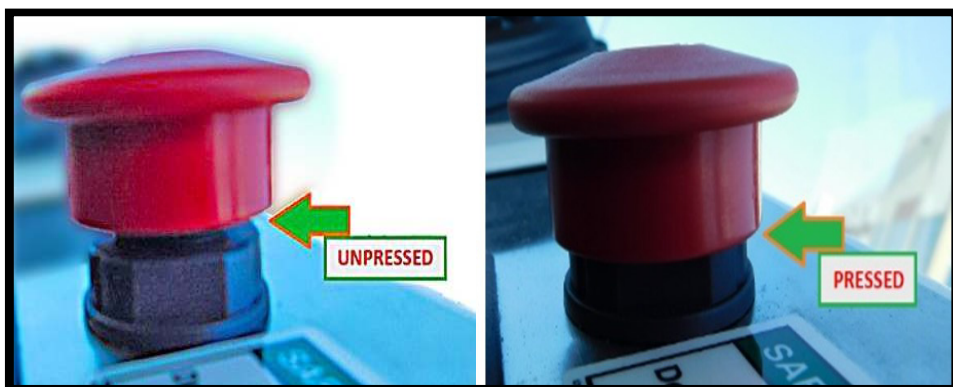
Pressing the SDHI does NOT lockout HV. Follow the process to disable the truck.

1. Disconnect from charger if connected.
2. Turn off 12 V battery disconnect. When in the off position, the 12 V disconnect can have a Lockout Tagout style or standard pad lock installed.



3. Turn off ignition.

4. Depress the SDHI Button.



5. Turn the key switch on.
6. If display screen powers on, High Voltage CANNOT be deenergized. DO NOT TOUCH any orange High Voltage cables. Contact OEV Service (866)-688-5223.
7. Turn the key switch off.
8. Remove key if applicable.



WARNING

The High Voltage system on this truck is NOT capable of being LOCKED OUT.

Always assume the high voltage system is energized. Do not touch any high voltage cables or components.

All steps must be followed to mitigate electrical hazard in an emergency.

4. Access to Occupants



Suspension Type Seats

The e-TRIEVER® is equipped with an air ride suspension seat.

WARNING

Due to the vertical travel of suspension seats, the operator must be sure that there is adequate head clearance when the seat is at the top of its upward travel.

Seat Adjustment

Adjust the seat to the comfortable position for the operator to have full control of all cab controls and displays. This position is obtained by changing the air pressure in the suspension, which changes the height. The seat height is controlled using a switch on the front-left side of the seat. The fore and aft position is obtained by moving the seat cushion forward and back on its slides.

Seat Heat

Your truck may be equipped with a heated seat. Press the heater control button located on the left hand side of the seat to engage the seat heating.



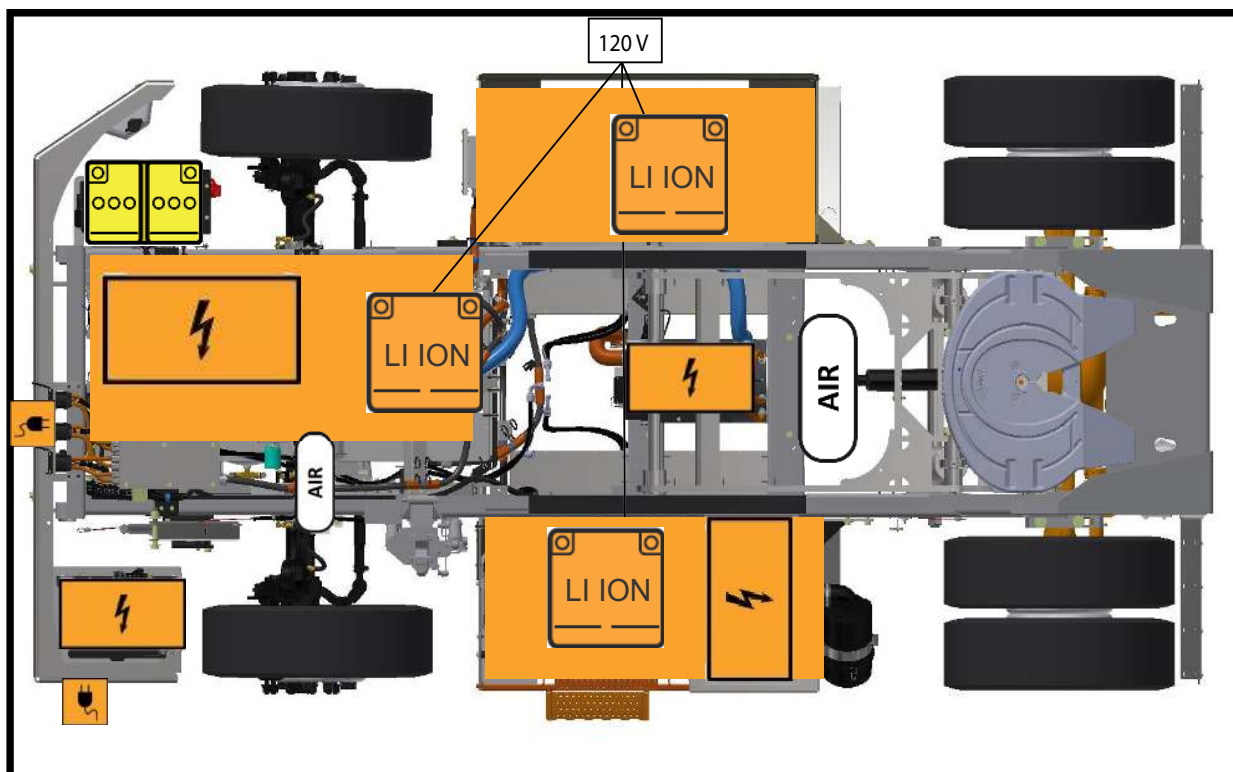
WARNING

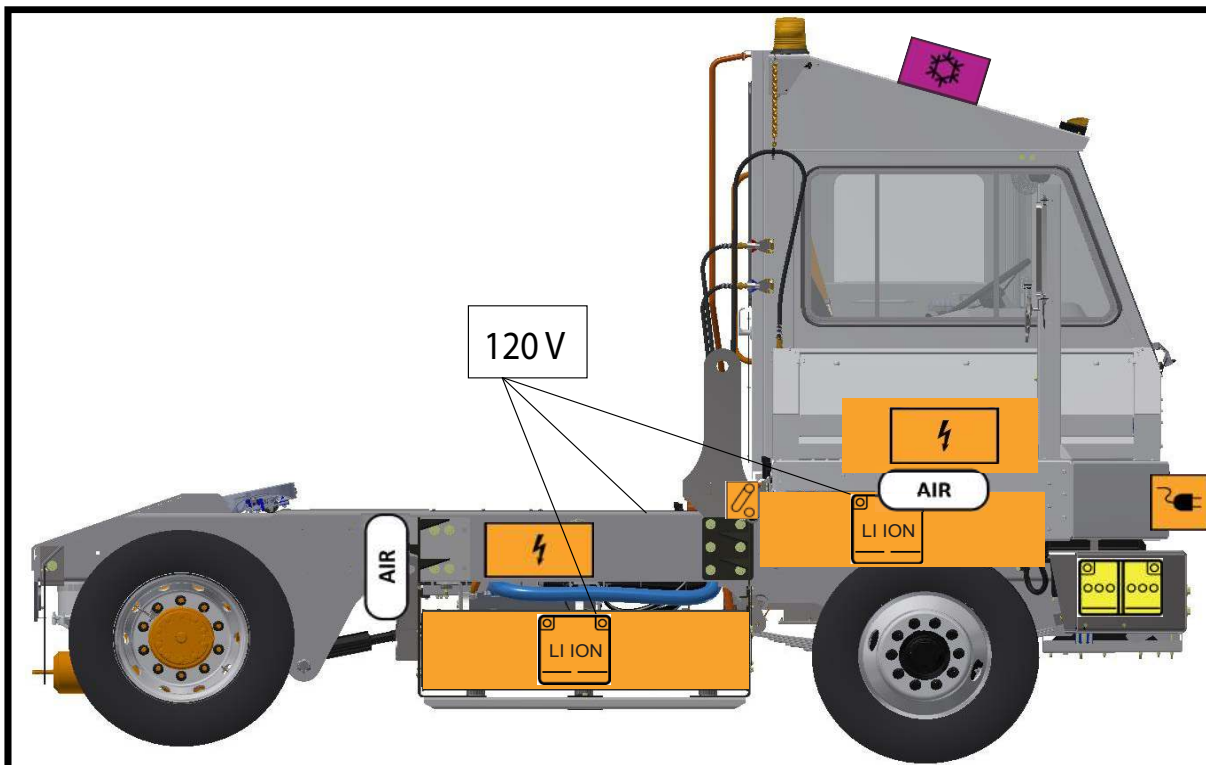
Do not adjust driver's seat unless truck is parked. The seat could move suddenly and unexpectedly and cause loss of control of the truck.

DANGER

The Orange EV truck may have a training seat installed. In this configuration the rear door may be partially blocked by seat and/or rider. Never allow anyone to ride on the outside of the cab. Carrying a passenger is extremely dangerous and can result in serious injury or death.

5. Stored Energy / Liquids / Gases / Solids





		12 V
		120 V



Never open high voltage components or cut orange high voltage cables. Structural PPE (Personal Protective Equipment) does not offer electrical protection.



The presence of a sweet, ether-like odor could be due to a leaking lithium-ion battery. Liquid on the ground around the truck is not an indicator of a battery leak. Lithium-ion electrolyte will form droplets, not puddles.

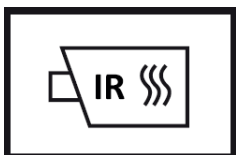


Damage to the pressurized hydraulic fluid system, or compressed air system could create high pressure fluid leaks and flying debris.

6. In Case of Fire



Use large, sustained quantities of water to extinguish any truck fire and cool down battery enclosures.



Use a thermal imaging camera to observe high voltage components. Ensure that the battery box has cooled sufficiently and is not increasing in temperature, indicating potential thermal runaway of the lithium-ion batteries. Monitor for hissing, popping, or smoke.



Avoid direct contact with high voltage battery assembly, cables, and components.



Batteries and electrolyte can release toxic fumes under fire conditions. Wear appropriate inhalation protection.

Lithium-ion batteries contain liquid, flammable electrolyte.

Burning batteries can also ignite other batteries in the vicinity.

The extinguishing agent must be applied continuously until fully cooled down; otherwise, there will be a risk of a new ignition.



- A burning lithium-ion battery generally cannot be extinguished directly. Water as the extinguishing agent can be used for cooling lithium-ion batteries.
- A battery fire may continue to burn for several hours or re-ignite, so it is recommended to continue to cool the battery with excessive amounts of water.
- The temperature of the battery can be monitored with an thermal imaging camera to ensure it is not heating up.
- Re-check the temperature of the battery once an hour to ensure the temperature is not increasing. If the temperature is increasing continue to cool the battery with water.

Do not store a truck containing a damaged or burning lithium-ion battery within 15 feet of a structure or other trucks.

7. In Case of Submersion



DANGER

Handling a submerged truck without appropriate training and personal protective equipment (PPE) can result in serious injury or death.

The removal and de-energizing of a partially or completely submerged truck should always be handled by trained emergency responders outfitted with the required PPE.

The high-voltage system of the e-TRIEVER® is isolated from the chassis. When undamaged, the system will not energize the surrounding water, even when fully submerged.

If you are in a truck when it is submerged, exit the truck if you can do so safely. To minimize risk, avoid contact with a submerged high voltage system and batteries. If there is time, and it is safe to do so, turn off the ignition before exiting the cab.

When you are in a safe location, immediately contact emergency services.

Emergency responders will check for damage and remove the truck from the water before disabling the electrical system using the steps outlines in “High Voltage Safety Device Hardware Interlock (SDHI) Activation” on page 8.



WARNING

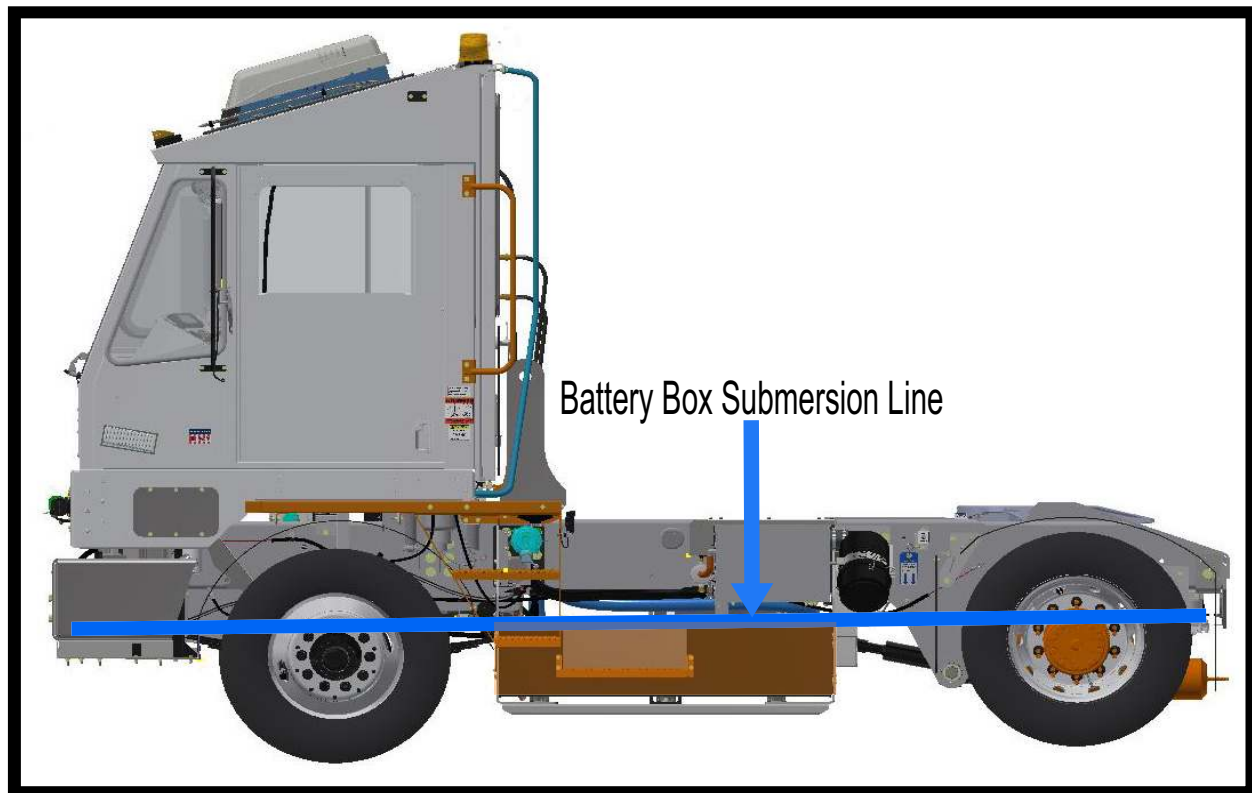
Only enter the water if properly equipped with PPE. Avoid contact with high voltage battery assembly, cables, and components.



WARNING

Submersion, especially in salt water, can damage components and create a fire risk.

The battery enclosures on e-TRIEVER® trucks are not fully sealed. If a truck is submerged such that the water level passes above the top of a battery box, the battery must be inspected by Orange EV before returning to service. Follow the storage instructions noted in “Storing the truck” on page 20 and contact OEV Service at (866) 688-5223.



8. Towing / Transportation / Storage



WARNING

Avoid contact with high voltage components and cables. In the case of a fire, relays could have fused closed due to heat, keeping the truck energized even after disabling.

Transporting the truck

Transporting a truck requires special equipment and training. Orange EV recommends that a professional transport service (or trained service professionals) be used to transport a disabled truck.

While the non-operating truck may be moved onsite a short distance with all six wheels on the ground, Orange EV recommends transporting the truck with all six wheels off the ground using a Landoll or comparable equipment. This avoids potential damage to the driveline or motor. This also avoids having to disconnect the rear axle and cage the rear spring brakes.

Towing truck with rear wheels suspended

If it is not possible to transport the truck with all six wheels off the ground, the secondary option is to tow the truck with the rear wheels suspended using either a professional transport service or trained service professional. The steering axle must be locked in the straight ahead position. If applicable, spring brakes on the front axle must be caged.

In a rare case where transport must occur with front wheels suspended, follow the steps listed in *Towing truck with front wheels suspended*.



Towing truck with front wheels suspended

It is not recommended to tow a truck with the front wheels lifted and the rear drive wheels on the road. This practice may result in serious truck damage. Pay close attention to the following rules to prevent truck damage if you must tow your Orange EV truck with the rear wheels on the road.



WARNING

Obey the following when towing an Orange EV truck with the front wheels lifted and the rear wheels on the road.

- 1. Use a rigid towing bar or properly restrain the towed truck. Using a chain or cable to tow the truck is not recommended.**
- 2. Disconnect the drive line from the rear axle.**
- 3. Cage the spring brakes on all rear drive wheels. (See section Manually Releasing Spring Brakes) Any loss of air pressure while towing the truck would apply the spring/parking brakes and lock the rear wheels on the towed truck.**

Storing the truck

Park the damaged truck a safe distance from combustible objects, buildings, and other vehicles until the battery can be properly discharged. Notify tow operator of stowing distance.



If damaged, a lithium-ion battery can increase in temperature over time and pose a delayed fire risk. Monitor battery temperature for 48 hours after incident using a thermal infrared camera.



9. Important Additional Information

The High Voltage System on this truck **CANNOT** be **LOCKED OUT**.

Never cut an orange cable.

Never cut a black cable with orange loom.



Do not perform any operation on a damaged truck without appropriate PPE.

Call OEV Service at (866) 688-5223 for assessment of truck post incident or for guidance about storing a damaged truck.

10. Explanation of Pictograms Used



Air Tank



Seat Height Adjustment by air system



Seat Adjustment, Longitudinal



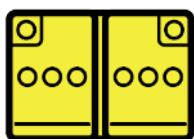
Air Conditioning Component



Steering Wheel, Tilt Control



Battery Pack, High-Voltage



Battery, Low Voltage



Warning, Electricity



Use Thermal Infrared Camera



Use Water to Extinguish the Fire



Acute Toxicity



Explosive



Corrosive



Flammable



Hazardous to Human Health



General Warning